

SFAX THYNA, Tunisia

WMO: 607500

Lat: 34.718N

Lon: 10.691E

Elev: 26

StdP: 101.01

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	5.8	-3.9	2.7	11.5	-2.1	3.2	11.2	11.1	13.9	10.0	13.9	2.4	300	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	36.1	22.6	34.2	22.7	33.0	22.7	26.1	31.0	25.4	30.4	24.8	29.8	4.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.0	29.0	24.0	18.9	28.5	23.1	18.0	28.2	81.0	31.2	78.0	30.3	75.4	29.9	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.4	7.4	1.8	41.7	0.9	2.7	1.2	43.6	0.6	45.2	0.1	46.8	-0.6	48.8		
(5)			0.2	27.3	1.2	1.0	-0.7	28.1	-1.4	28.7	-2.1	29.2	-2.9	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	12.2	12.7	15.1	17.8	21.4	25.1	27.7	28.3	26.0	22.6	17.3	13.5	(6)
(7)		DBStd	6.16	1.94	2.30	2.60	2.38	2.39	2.58	1.96	1.83	2.28	2.57	2.65	2.26	(7)
(8)		HDD10.0	9	4	3	1	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD18.3	693	189	158	106	36	3	0	0	0	0	1	49	151	(9)
(10)		CDD10.0	3669	73	79	158	236	352	452	550	568	481	391	220	111	(10)
(11)		CDD18.3	1311	0	1	6	22	96	202	292	309	231	134	19	1	(11)
(12)		CDH23.3	13201	1	9	52	114	583	1909	3515	3834	2239	859	80	4	(12)
(13)	CDH26.7	5266	0	1	13	21	147	717	1612	1793	763	192	7	0	(13)	
(14)	Wind	WSAvg	3.7	3.5	3.7	3.8	4.2	4.1	3.9	3.7	3.5	3.6	3.3	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	225	23	20	26	19	9	4	1	4	24	46	21	27	(15)
(16)		PrecMax	652	183	124	65	104	64	39	15	32	117	463	112	152	(16)
(17)		PrecMin	91	0	0	1	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	112	32	24	18	23	12	7	3	8	27	80	24	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	24.4	28.0	29.2	33.2	39.0	39.7	38.6	35.3	33.1	27.2	23.3	(19)
(20)			MCWB	12.9	14.2	14.9	17.1	19.4	21.1	22.5	23.0	23.3	20.8	17.4	13.9	(20)
(21)		2%	DB	20.0	21.4	24.2	26.2	30.0	34.1	35.8	36.0	33.1	30.1	25.2	21.1	(21)
(22)			MCWB	12.1	12.9	14.7	16.4	19.0	21.0	22.3	23.1	23.8	21.2	17.7	13.6	(22)
(23)		5%	DB	18.2	19.4	22.1	24.1	28.0	32.0	34.0	34.2	31.6	28.8	24.0	19.9	(23)
(24)			MCWB	11.8	12.4	14.2	15.8	18.7	21.0	22.4	23.5	23.3	21.3	17.3	13.4	(24)
(25)		10%	DB	17.0	17.9	20.2	22.9	26.2	30.2	32.6	33.0	30.2	27.2	22.8	18.6	(25)
(26)			MCWB	11.6	11.9	13.8	15.4	18.1	20.8	22.3	23.5	22.9	20.9	16.5	13.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	15.6	17.6	19.4	22.1	25.3	26.4	27.7	26.7	25.0	21.1	16.9	(27)
(28)			MCDWB	18.0	20.3	23.1	25.4	28.3	31.7	31.7	32.7	30.7	28.0	24.0	19.4	(28)
(29)		2%	WB	13.9	14.4	16.4	18.0	20.9	23.7	25.5	26.4	25.7	23.8	19.8	15.6	(29)
(30)			MCDWB	17.2	18.8	21.0	23.2	26.7	29.7	30.8	31.5	29.6	27.1	23.2	18.6	(30)
(31)		5%	WB	13.1	13.6	15.5	17.2	20.1	22.7	24.7	25.6	25.1	22.9	18.6	14.7	(31)
(32)			MCDWB	16.6	17.8	19.9	21.8	25.3	28.8	30.4	30.9	29.1	26.4	22.3	18.0	(32)
(33)		10%	WB	12.3	12.8	14.7	16.5	19.3	21.9	24.0	25.0	24.4	22.1	17.5	13.7	(33)
(34)			MCDWB	16.1	16.9	18.8	21.0	24.6	28.0	29.9	30.4	28.7	25.9	21.3	17.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	10.2	10.1	9.3	9.2	9.6	9.6	9.2	8.2	8.8	10.0	10.0	(35)
(36)			MCDBR	12.2	12.9	13.0	11.9	11.6	12.3	11.8	11.4	9.4	10.2	10.9	11.5	(36)
(37)			MCWBR	6.9	6.7	6.1	5.3	4.9	5.0	5.1	4.9	4.4	4.9	5.9	6.5	(37)
(38)		5% WB	MCDBR	10.1	11.0	10.5	9.7	9.7	10.6	9.9	9.6	8.3	8.7	9.1	9.5	(38)
(39)			MCWBR	6.5	6.4	5.6	4.9	4.7	5.1	4.7	4.7	4.3	4.6	5.8	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.356	0.383	0.458	0.509	0.522	0.485	0.511	0.502	0.505	0.461	0.407	0.368	(40)
(41)		taud		2.356	2.258	2.032	1.929	1.925	2.057	2.007	2.037	2.029	2.103	2.205	2.307	(41)
(42)		Ebn at Noon		837	851	810	788	781	809	785	784	758	755	763	793	(42)
(43)		Edh at Noon		97	119	162	188	191	167	175	167	161	137	112	97	(43)
(44)	All-Sky Solar Radiation	RadAvg		2.95	3.83	5.01	6.09	7.07	7.62	7.74	6.84	5.39	4.15	3.15	2.64	(44)
(45)		RadStd		0.15	0.20	0.26	0.31	0.32	0.29	0.18	0.23	0.20	0.21	0.12	0.17	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.21	N/A	N/A	N/A	-0.56	-0.75	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.24	N/A	N/A	N/A	-0.37	-0.33	N/A	N/A	+111	+58

Nomenclature: See separate page